Submission Worksheet

Submission Data

Course: IT202-450-M2025
Assignment: IT202 PHP Multi-Dimension Problems
Student: Wilfer G. (wg99)
Status: Submitted | **Worksheet Progress:** 100%
Potential Grade: 1000.00/1000.00 (100.00%)
Received Grade: 0.00/1000.00 (0.00%)
Started: 7/14/2025 12:30:47 PM
Updated: 7/14/2025 12:55:24 PM
Grading Link: https://learn.ethereallab.app/assignment/v3/IT202-450-M2025/it202-php-multi-dimension-problems/grading/wg99
View Link: https://learn.ethereallab.app/assignment/v3/IT202-450-M2025/it202-php-multi-dimension-problems/view/wg99

Instructions

- Overview Link: https://youtu.be/jd5giyXReaI

1. Ensure you read all instructions and objectives before starting.
2. Create a new branch from `dev` called `M6-Homework`
    1. `git checkout dev` (ensure proper starting branch)
    2. `git pull origin dev` (ensure history is up to date)
    3. `git checkout -b M6-Homework` (create and switch to branch)
3. Copy the template code from here: GitHub Repository - M6 Homework
    - It includes Problems 1-3 and `base.php`. Put all into an `M6` folder or similar inside your `public_html`
    - Immediately record to history
        - `git add public_html`
        - `git commit -m "adding M6 HW baseline files"`
        - `git push origin M6-Homework`
        - Create a Pull Request from `M6-Homework` to `dev` and keep it open
4. Fill out the below worksheet
    - Each Problem requires the following as you work
        - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
        - Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
        - Code solution (add/commit periodically as needed)
5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
    1. `git add .`
    2. `git commit -m "adding PDF"`
    3. `git push origin M6-Homework`
    4. On Github merge the pull request from `M6-Homework` to `dev`
    5. On Github create a pull request from `dev` to `prod` and immediately merge. (This will trigger the `prod` deploy to make the heroku prod links work)

7. Upload the same PDF to Canvas
8. Sync Local
    1. `git checkout dev`
    2. `git pull origin dev`

Section #1: (250 pts.) Problem 1 - Subset

Progress: 100%

☰ Task #1 (250 pts.) - Edit the `processBirds` function to extract properties

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Extract the name, color, region of each bird into the $subset array
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

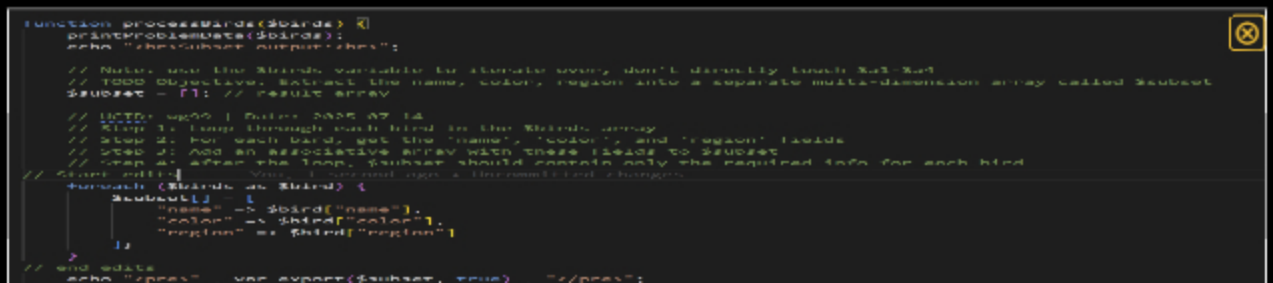
🖼 Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Heroku dev after a manual deploy)

the output



```
$myApi = "wg99nj"; // replace with your UCSB
printHeader($myApi, 1);
```

the code

Saved: 7/14/2025 12:39:45 PM

Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

URL #1
https://it202-738691a16289.herokuapp.com/M6/problem1.php

URL
https://it202-738691a16289.herol

URL #2
https://github.com/wg99nj/IT202-2025M6---Homework/public_html/M6/problem1.php

URL
https://github.com/wg99nj/IT202-

Saved: 7/14/2025 12:39:45 PM

Part 3:

Progress: 100%

Details:
Briefly explain `how` the code solves the challenge (note: this isn't the same as `what` the code does)

Your Response:

The code solves the challenge by looping through each bird and creating a new array that only includes the name, color, and region for each one. This keeps things simple and makes sure you only get the details you need for the assignment.

Saved: 7/14/2025 12:39:45 PM

Section #2: (250 pts.) Problem 2 - Adding Properties

Progress: 100%

Task #1 (250 pts.) - Edit the `processCars` function to add

properties

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Add a new property called age that's set from today's year and the car's year
- Challenge 2: Add a new property called isClassic that's true/false based on $classic_age
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Heroku dev after a manual deploy)

the output

```
function processCars($cars) {
    printProblemData($cars);
    echo "<h1>New Properties Output:</h1>";

    // Note: use the $cars variable to iterate over, don't directly touch $a1-$a4
    // TODO Objective: Add $age to create a new array ($processedCars) with original properties plus age and isClassic. isClassic is a boolean
    $currentYear = (int)date("Y"); // determine current year
    $processedCars = []; // result array
    $classic_age = 25; // don't change this value

    // UCID: wg99 | Date: 2025-07-14
    // Step 1: Loop through each car in the $cars array
    // Step 2: For each car, calculate age as currentYear - car's year
    // Step 3: For each car, set isClassic to true if age >= $classic_age, else false
    // Step 4: Add all original properties plus age and isClassic to $processedCars

    // Start edits
    foreach ($cars as $car) {
        $age = $currentYear - $car["year"];
        $isClassic = $age >= $classic_age;
        $car["age"] = $age;
        $car["isClassic"] = $isClassic;
        $processedCars[] = $car;
    }
    // End edits
    echo "<pre>" . var_export($processedCars, true) . "</pre>";
}
```

the code

Saved: 7/14/2025 12:41:36 PM

Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

URL #1
https://it202-738691a16289.herokuapp.com/M6/problem2.php

URL
https://it202-738691a16289.herol

URL #2
https://github.com/wg99nj/IT202-DOS-M6---Homework/public_html/M6/problem2.php

URL
https://github.com/wg99nj/IT202-

Saved: 7/14/2025 12:41:36 PM

Part 3:

Progress: 100%

Details:
Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

The code solves the challenges by automatically calculating each car's age based on the current year and then determining if the car qualifies as a classic using a set threshold. By adding these new properties to each car, the solution makes it easy to identify which cars are considered classics and how old each car is, without needing to manually check or calculate these values outside the code. This approach streamlines the process and ensures the results are always up-to-date.

Saved: 7/14/2025 12:41:36 PM

Section #3: (250 pts.) Problem 3 - Join

Progress: 100%

Task #1 (250 pts.) - Edit the `joinArrays` function to combine two arrays based on a common key

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Combine the data in both arrays by the userId property

- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Heroku dev after a manual deploy)

the output

the code

Saved: 7/14/2025 12:48:09 PM

Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`)
- Direct link to the file on Heroku Prod (Just grab the base prod url and manually enter the path to the file)

URL #1

https://it202-738691a16289.herokuapp.com/M6/problem3.php

https://it202-738691a16289.herol

URL #2

https://github.com/wg99nj/IT202-M6---Homework/public_html/M6/problem3.php

https://github.com/wg99nj/IT202-

Saved: 7/14/2025 12:48:09 PM

Part 3:

Progress: 100%

Details:

Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

The code solves the challenge by automatically matching each user with their activity using the userId, so you get a single combined record for each person. This makes it easy to see both the user's details and their activity together, without having to manually look up or pair the information from separate arrays.

Saved: 7/14/2025 12:48:09 PM

Section #4: (250 pts.) Misc

Progress: 100%

Task #1 (83.33 pts.) - Github Details

Progress: 100%

Part 1:

Progress: 100%

Details:

From the Commits tab of the Pull Request screenshot the commit history Following minimum should be present

Files

1 hr 41 mins	lib/config.php
1 hr 26 mins	public_html/Project/register.php
1 hr 18 mins	public_html/Project/profile.php
48 mins	public_html/Project/login.php
40 mins	...ic_html/Project/admin/create_role.php
38 mins	...ql/002_alter_table_users_username.sql
36 mins	lib/db.php
33 mins	lib/duplicate_user_details.php
32 mins	public_html/Project/sql/init_db.php
29 mins	.env
23 mins	public_html/Project/index.php
20 mins	...roject/sql/001_create_table_users.sql
18 mins	lib/reset_session.php
18 mins	public_html/Project/styles.css

Branches

11 hrs 26 mins	Milestone1
47 mins	M6---Homework
8 mins	dev

commit history

Saved: 7/14/2025 12:55:24 PM

Part 2:

Progress: 100%

Details:
Include the link to the Pull Request (should end in `/pull/#`)

URL #1
https://github.com/wg99nj/IT202-2025/pull/17

URL
https://github.com/wg99nj/IT202-

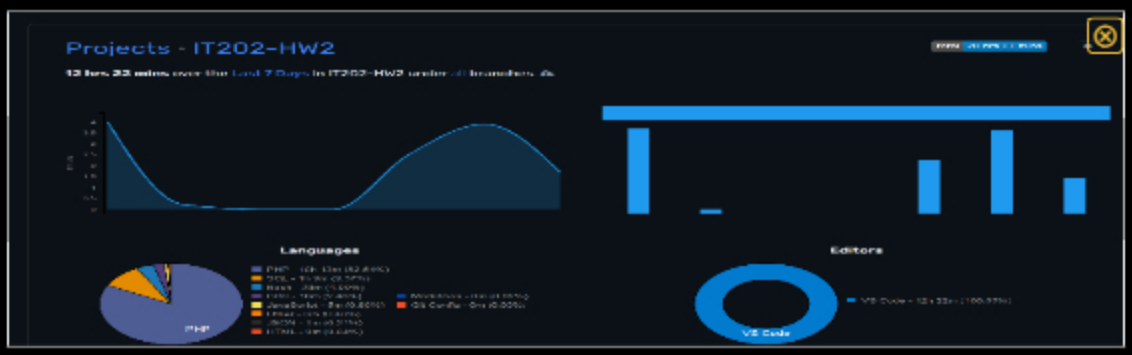
Saved: 7/14/2025 12:55:24 PM

Task #2 (83.33 pts.) - WakaTime - Activity

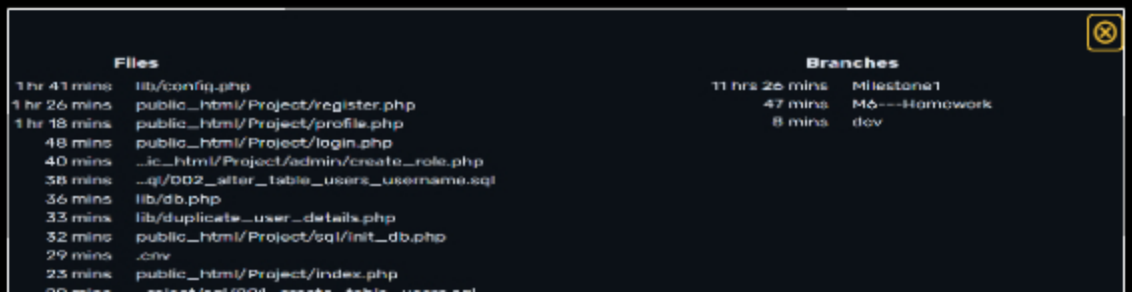
Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click `Projects` and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary



wakatime



20 mins ...roject/sql/001_create_table_users.sql
18 mins lib/reset_session.php
18 mins public_html/Project/styles.css

time duration

Saved: 7/14/2025 12:51:08 PM

Task #3 (83.33 pts.) - Reflection

Progress: 100%

Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

In this challenge, I learned how to join two arrays based on a common property, such as userId, to create a combined dataset. I practiced using nested loops and array functions in PHP to efficiently match and merge related records. This helped me understand how to organize and connect data from different sources.

Saved: 7/14/2025 12:51:54 PM

Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

The easiest part of the assignment was setting up the initial arrays and writing the loop to combine the user and activity data. Since the data was already organized and the userId property was consistent across both arrays, it was straightforward to match and merge the records.

Saved: 7/14/2025 12:52:52 PM

Task #3 (0.33 pts.) - What was the hardest part of the

assignment?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

The hardest part of the assignment was figuring out how to efficiently match and combine the data from the two arrays using the userId property. It required careful looping and logic to make sure each user was paired with the correct activity, and to avoid mistakes like mismatched or missing data.

Saved: 7/14/2025 12:53:37 PM